OligoN-design:
A simple and versatile tool to design specific probes and primers from large heterogeneous datasets

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Motivation

In the big data era, designing taxon-specific oligonucleotides is a current **bottleneck** of different molecular studies that rely on primers for PCR or probes for FISH.

Existing tools rely on specific databases, alignments or phylogenetic trees, and/or cannot accommodate increasingly large molecular environmental datasets.

Here we present **oligoN-design**, a versatile tool to design oligonucleotides specific to a set of target sequences while minimizing predicted binding to non-target sequences.

Novelty

OligoN-design is a collection of functions to ease oligonucleotide design.

It requires only two fasta files as input:

Target file: Contains the target taxa & **Excluding file**: Contains non-target taxa

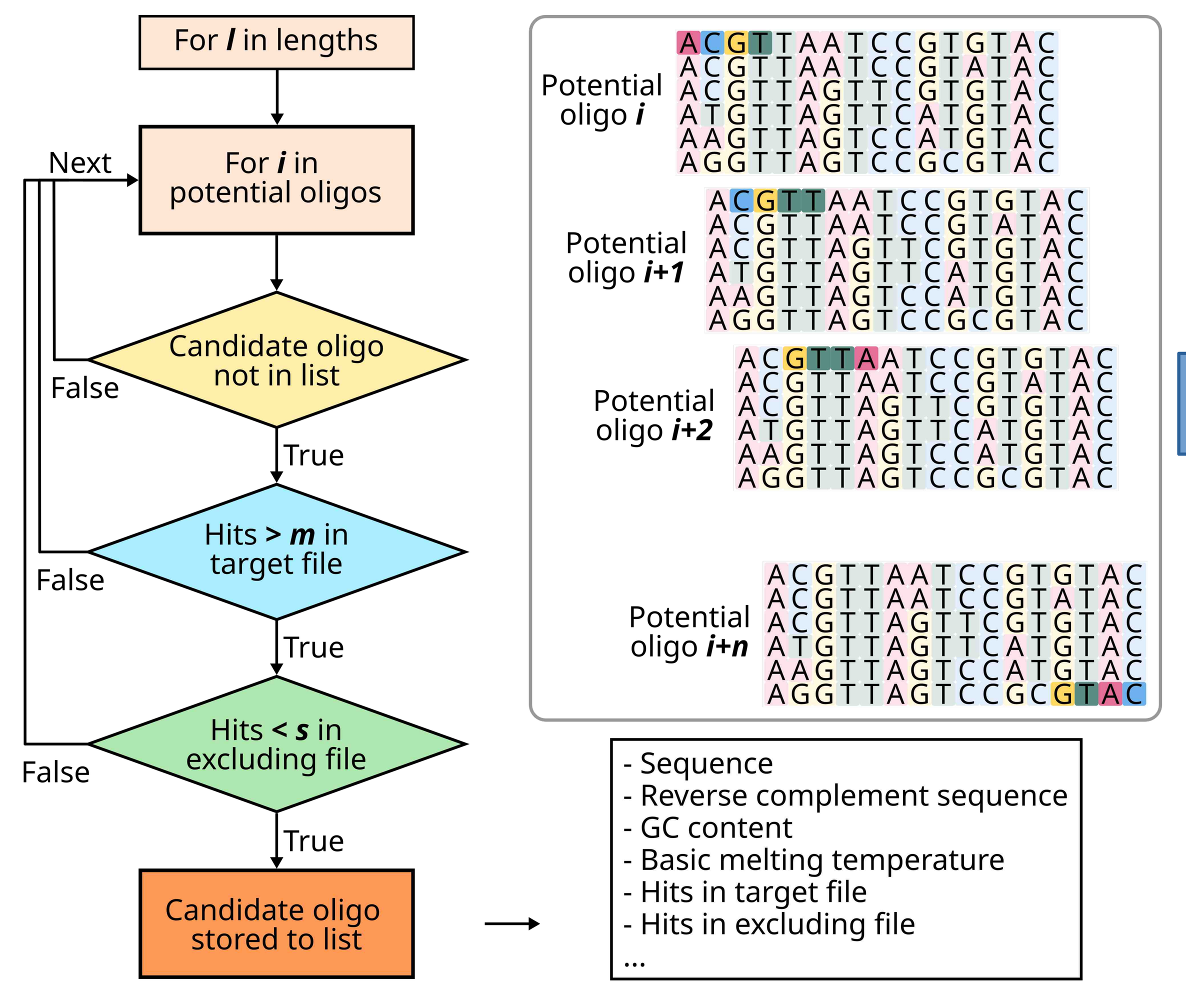
It has been designed to be **simple**, **versatile**, **reproducible** to **accommodate large and heterogeneous datasets,** and does not require phylogenetic trees or alignments.

OligoN-design in a nutshell

1) Find oligos

Searches candidate oligonucleotides by sliding windows:

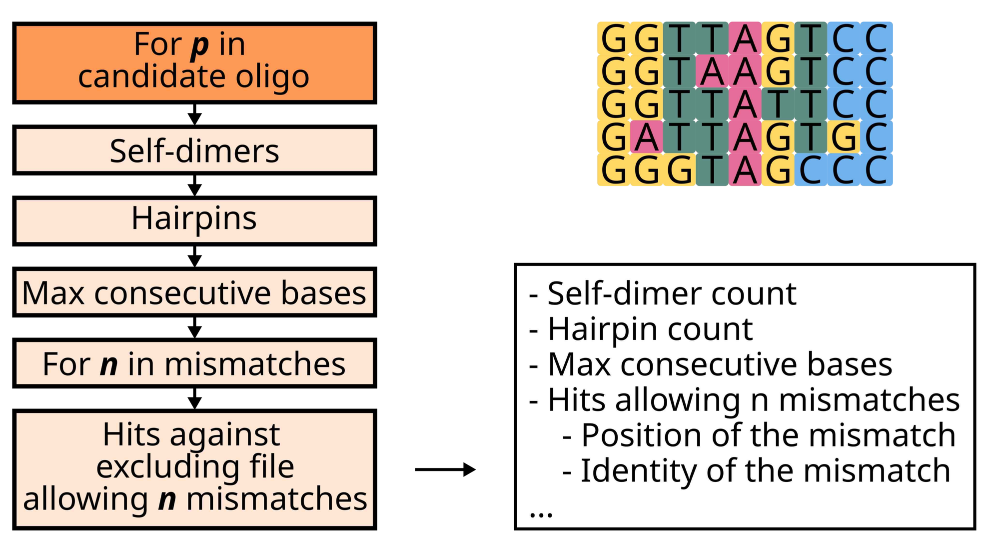
`findOligo -t target -e excluding -l length -s specificity -m match`



2) Test oligos

A quick test for mismatches

`testOligo -f primers -e excluding`



and/or

A thorough test for self-dimers, hairpin, mismatches and their position in the oligo, ...

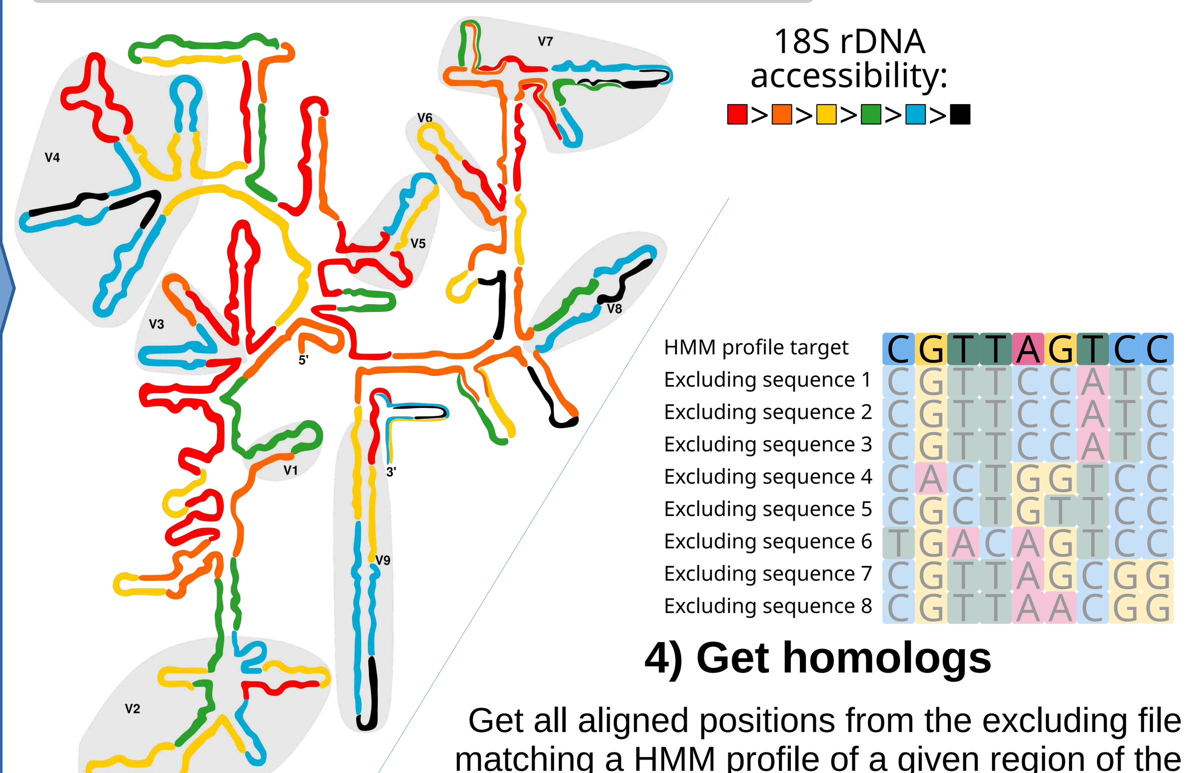
`testThorough -f oligos -e excluding`

3) Rate accessibility

Compare candidate oligonucleotides to the accessibility map of *Saccharomyces cerevisae* 18S rDNA or *Escherichia coli* 16S (Behrens et al., 2003; *Appl. Environ. Microbiol.*)

`alignPrimers -t target -p oligos -o oligos_align`

`rateAccess -f oligos_align -o oligos_access`



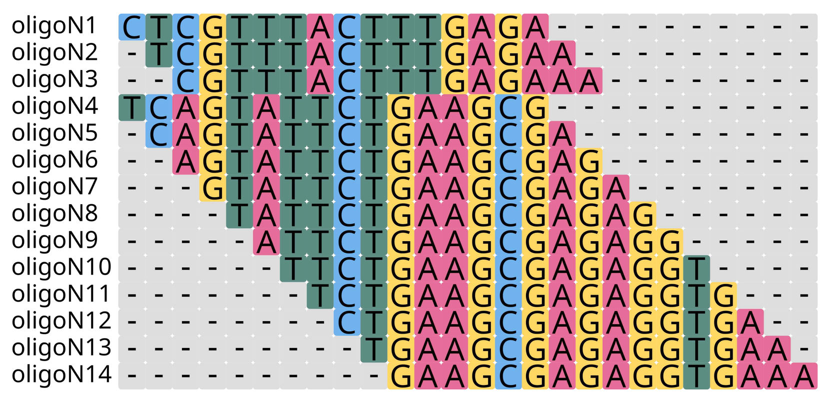
4) Get homologs

Get all aligned positions from the excluding file matching a HMM profile of a given region of the target file

`getHomolog -f region -e excluding`

5) Identify regions

From a fasta file containing different potential oligonucleotides, identifies regions of interest and groups together oligonucleotides related to the same region.

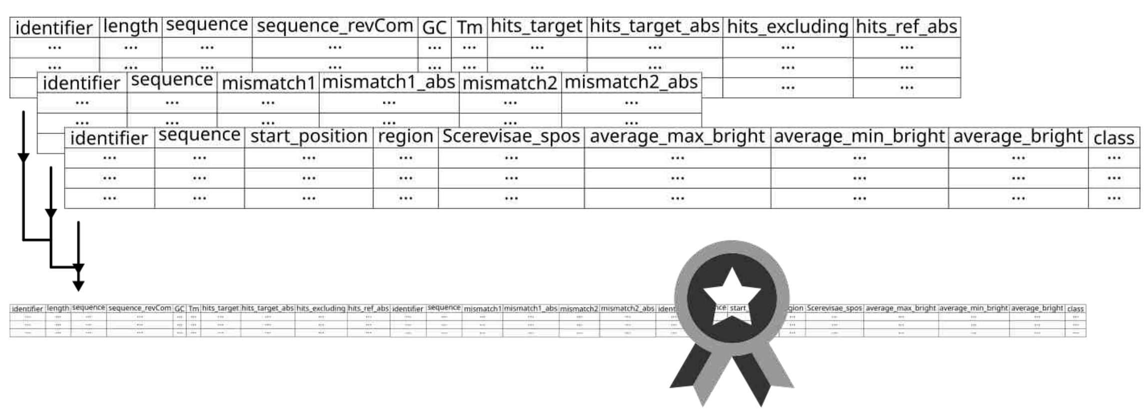


`identifyRegions -f oligos_align -e oligos_regions`

6) Selecting oligonucleotides

Merge all log files:

`bindLogs -f oligos oligos_tested oligos_access -o oligos_log -r`



Selecting the best oligonucleotide (e.g.):

`filterLog -l oligos_log -s 0.4 -m 0.001 -M 0.0001 -c III`

`selectLog -l oligos_log_filtered -n 4`

Usage and performance

Using large, comprehensive ribosomal databases (i.e., PR2, SILVA, eukRibo) and an unsupervised function, we were able to replicate known taxa-specific oligonucleotides in under 30 minutes and up to 6 Gb of RAM on a personal laptop.

See the manual for further details!

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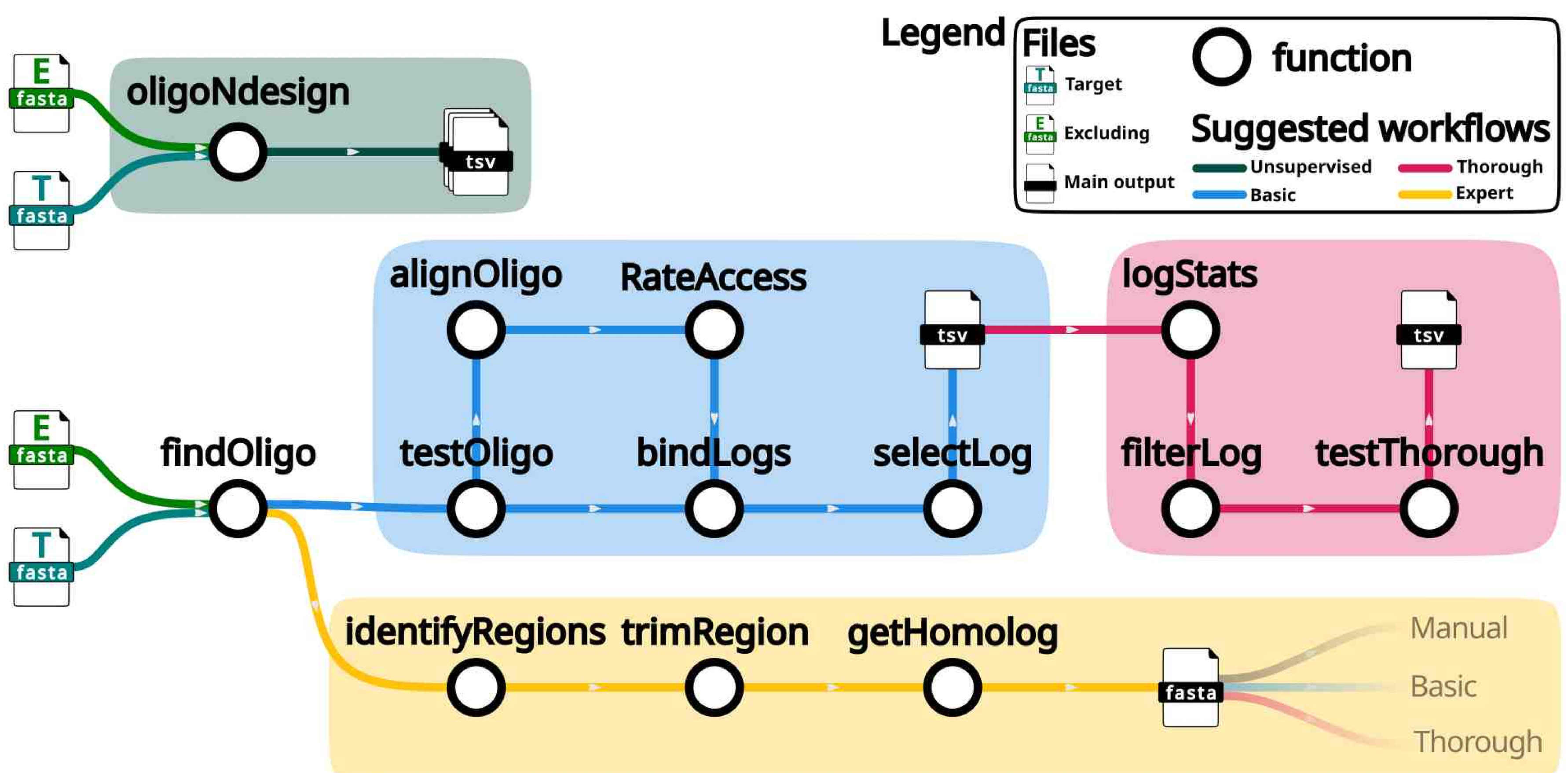
Integration and workflows

OligoN-design allows a range of strategies, from an unsupervised end-to-end usage all the way to a detailed and thorough expert usage.



Where to find OligoN-design

OligoN-design is available at **github.com/MiguelMSandin/oligoN-design**

under GNU General Public License version 3.0,

can be easily installed via **bioconda**,

`mamba install oligon-design`

and comes with a fully detailed usage manual.

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OligoN Design

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